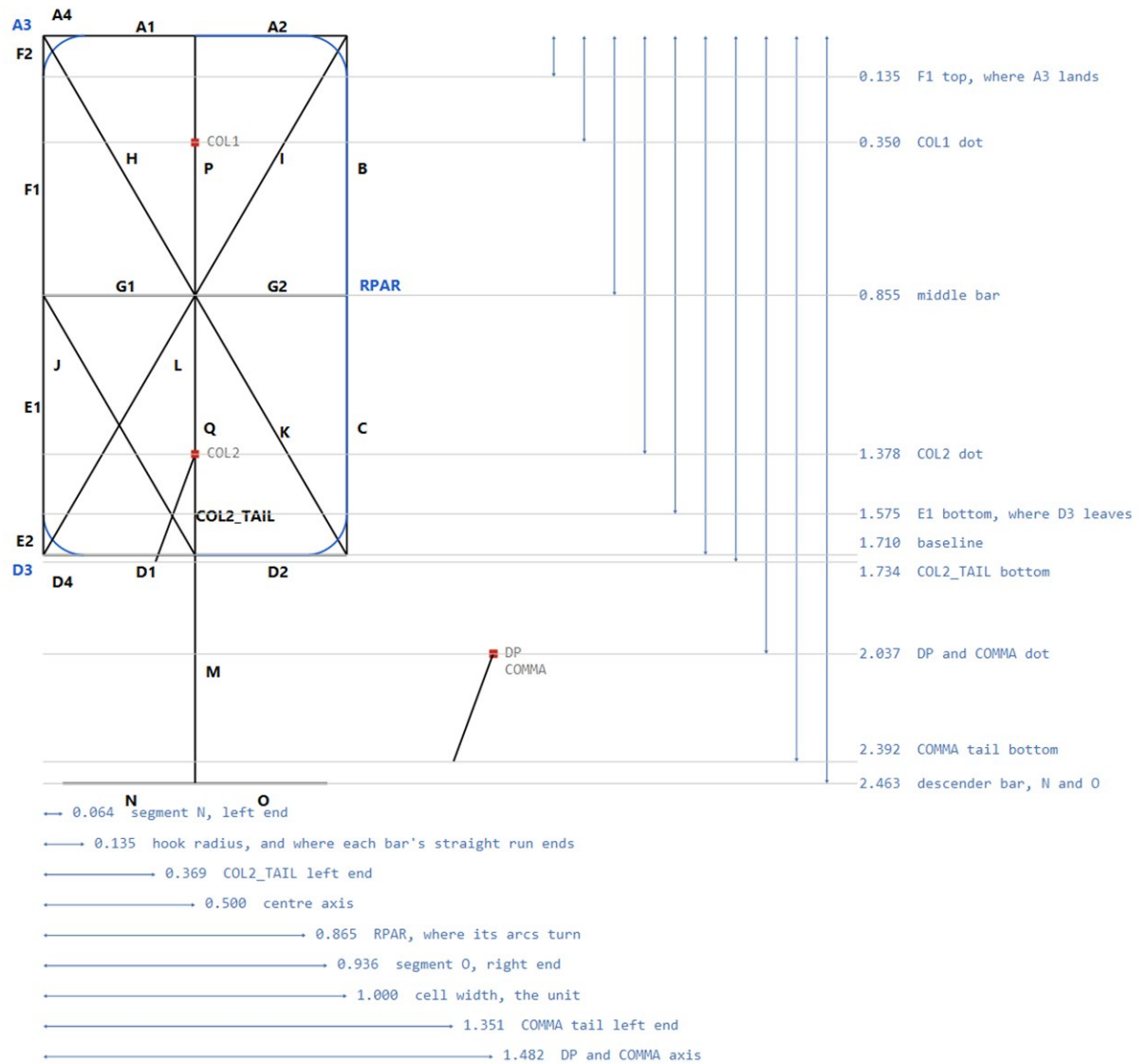


## TalkRPN cell - centreline skeleton

Inspired by the HP-01 (1977) and the Litronix DL-3422, and identical to neither. The HP-01's look - hooked corners, the slant, the proportions - carried onto roughly the DL-3422's segment count, so the display can show letters as well as digits. Departures from both are deliberate: the corners hook where the DL-3422's are square, the segments meet flush where a real part leaves gaps, the decimal point sits between cells rather than taking one of its own, and there are elements here that neither part had. It is a period style, not a reproduction of any real component.

THE UNIT: segment E/F to segment B/C is 1 - the left column to the right column, centre to centre. Every length here is in that unit, across and down alike, so the centre axis is exactly 0.5 and pitch minus 1 is the clearance between neighbouring cells.

No stroke width, no slant: these are centrelines, and both are applied at render time. Origin is the top-left centreline corner; x right, y down. 28 bars + COL1 + COL2 + DP + COMMA = 32 elements; the mask is a long. Blue: A3/D3 are ALTERNATIVES to A4/D4; RPAR is the whole right paren. TalkRpnFont.kt is the source of truth - this sheet mirrors it.



## TalkRPN cell - centreline listing

| SEGMENT   | CENTRELINES | from         | to                                                  |                               |
|-----------|-------------|--------------|-----------------------------------------------------|-------------------------------|
| A1        | line        | 0.500, 0.000 | -> 0.135, 0.000                                     |                               |
| A2        | line        | 0.500, 0.000 | -> 1.000, 0.000                                     |                               |
| A4        | line        | 0.135, 0.000 | -> 0.000, 0.000                                     |                               |
| A3        | arc         | 0.135, 0.000 | -> 0.000, 0.135                                     | arc R0.135 centre 0.135,0.135 |
| B         | line        | 1.000, 0.000 | -> 1.000, 0.855                                     |                               |
| C         | line        | 1.000, 0.855 | -> 1.000, 1.710                                     |                               |
| D1        | line        | 0.500, 1.710 | -> 0.135, 1.710                                     |                               |
| D2        | line        | 0.500, 1.710 | -> 1.000, 1.710                                     |                               |
| D4        | line        | 0.135, 1.710 | -> 0.000, 1.710                                     |                               |
| D3        | arc         | 0.135, 1.710 | -> 0.000, 1.575                                     | arc R0.135 centre 0.135,1.575 |
| F2        | line        | 0.000, 0.000 | -> 0.000, 0.135                                     |                               |
| F1        | line        | 0.000, 0.135 | -> 0.000, 0.855                                     |                               |
| E1        | line        | 0.000, 0.855 | -> 0.000, 1.575                                     |                               |
| E2        | line        | 0.000, 1.575 | -> 0.000, 1.710                                     |                               |
| G1        | line        | 0.000, 0.855 | -> 0.500, 0.855                                     |                               |
| G2        | line        | 0.500, 0.855 | -> 1.000, 0.855                                     |                               |
| H         | line        | 0.000, 0.000 | -> 0.500, 0.855                                     |                               |
| I         | line        | 1.000, 0.000 | -> 0.500, 0.855                                     |                               |
| J         | line        | 0.000, 0.855 | -> 0.500, 1.710                                     |                               |
| L         | line        | 0.500, 0.855 | -> 0.000, 1.710                                     |                               |
| K         | line        | 0.500, 0.855 | -> 1.000, 1.710                                     |                               |
| P         | line        | 0.500, 0.000 | -> 0.500, 0.855                                     |                               |
| Q         | line        | 0.500, 0.855 | -> 0.500, 1.710                                     |                               |
| COL2_TAIL | line        | 0.500, 1.378 | -> 0.369, 1.734                                     |                               |
| M         | line        | 0.500, 1.710 | -> 0.500, 2.463                                     |                               |
| N         | line        | 0.064, 2.463 | -> 0.500, 2.463                                     |                               |
| O         | line        | 0.500, 2.463 | -> 0.936, 2.463                                     |                               |
| RPAR      | compound    | 0.500,0.000  | -> arc -> 1.000, 0.135..1.575 -> arc -> 0.500,1.710 | the whole right paren         |

DOT CENTRES

|       |              |                                              |
|-------|--------------|----------------------------------------------|
| COL1  | 0.500, 0.350 | upper colon dot; also dots i and j           |
| COL2  | 0.500, 1.378 | lower colon dot, colon only                  |
| DP    | 1.482, 2.037 | decimal point, outside the cell to the right |
| COMMA | 1.482, 2.037 | same centre, plus a tail down and left       |

SQUARE, side = 2 x STROKE = 0.295 - a real HP-55's decimal point is a square die

CENTRELINE BOX 1.000 wide x 1.710 tall aspect 0.585  
 2.463 tall including the descender  
 PITCH 2.430 origin to origin; minus 1 IS the clearance between cells  
 VPITCH floor is 2.610, the ink height - below that descenders reach the next row  
 SLANT 6.0 degrees, applied at render

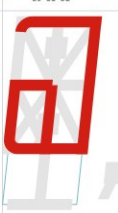
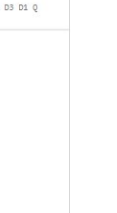
### RELATIONSHIPS

hook radius = hook start x = 0.135, so each arc is tangent to both bar and column  
 F1 top = 0.135 = where A3 lands. E1 bottom = 1.575 = where D3 lands.  
 F2 and E2 are the stubs that carry the left column out to the corner when it is square.  
 B meets C at y = 0.855: the right column is undivided at the corners, which is what makes a 4 lopsided.  
 N and O are inset 0.0640 and 0.0641 from the columns - symmetric to within measurement.  
 COL2\_TAIL and the COMMA tail are the SAME tail, 0.355 down and 0.131 left, hung off two different dots.  
 That is what makes a semicolon a colon whose lower dot grew a tail, at no cost in new geometry.

### THE THREE LEFT-COLUMN ENDINGS

|        |          |                                                                  |
|--------|----------|------------------------------------------------------------------|
| square | A4 + F2  | most letters                                                     |
| hooked | A3 + F1  | 0 2 3 5 7 8 9, A C G O Q S, ( & and @ (F2 dark)                  |
| short  | F1 alone | digit 4, no A at all - left side sits 0.135 lower than the right |



|   | 0                                                                                            | 1                                                                                        | 2                                                                                        | 3                                                                                        | 4                                                                                        | 5                                                                                        | 6                                                                                        | 7                                                                                         | 8                                                                                          | 9                                                                                          | A                                                                                          | B                                                                                          | C                                                                                          | D                                                                                          | E                                                                                          | F                                                                                          |
|---|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 2 | <br>sp       | <br>!   | <br>"   | <br>#   | <br>\$  | <br>%   | <br>&   | <br>'   | <br>(   | <br>)   | <br>*   | <br>+   | <br>,   | <br>-   | <br>.   | <br>/   |
| 3 | <br>0        | <br>1   | <br>2   | <br>3   | <br>4   | <br>5   | <br>6   | <br>7   | <br>8   | <br>9   | <br>:   | <br>;   | <br><   | <br>=   | <br>>   | <br>?   |
| 4 | <br>@        | <br>A   | <br>B   | <br>C   | <br>D   | <br>E   | <br>F   | <br>G   | <br>H   | <br>I   | <br>J   | <br>K   | <br>L   | <br>M   | <br>N   | <br>O   |
| 5 | <br>P       | <br>Q  | <br>R  | <br>S  | <br>T  | <br>U  | <br>V  | <br>W  | <br>X  | <br>Y  | <br>Z  | <br>[  | <br>\  | <br>]  | <br>^  | <br>_  |
| 6 | <br>\<br>h | <br>a | <br>b | <br>c | <br>d | <br>e | <br>f | <br>g | <br>h | <br>i | <br>j | <br>k | <br>l | <br>m | <br>n | <br>o |
| 7 | <br>p      | <br>q | <br>r | <br>s | <br>t | <br>u | <br>v | <br>w | <br>x | <br>y | <br>z | <br>[ | <br>] | <br>[ | <br>] | <br>] |